- **Natural history:** Hyperfiltration (early ↑ GFR) → microalbuminuria (30–300 mg/g) → macroalbuminuria (>300 mg/g) → progressive GFR decline → ESRD.
- **Classic presentation:** Long-standing diabetes (typically ≥10 years for type 1; variable for type 2) + albuminuria + retinopathy.

Diagnosis and When to Biopsy

When to Consider Kidney Biopsy (Atypical Features):

- **Absence of retinopathy** (present in >90% of type 1 DM with nephropathy; less reliable in type 2).
- **Rapid decline in GFR** or **sudden onset of heavy proteinuria**.
- **Active urine sediment** (RBC casts, dysmorphic RBCs) suggesting glomerulonephritis.
- **Short duration of diabetes** (<5 years for type 1) with significant proteinuria.
- **Systemic symptoms** suggesting alternative diagnosis (rash, arthritis, etc.).

Management

Core Principles for DKD Management:

- **ACEi or ARB:** First-line for patients with diabetes + albuminuria (*slows progression to ESRD*). Monitor Cr and K within 1–2 weeks of initiation.
- **SGLT2 Inhibitors (empagliflozin, dapagliflozin, canagliflozin):** Proven to reduce CKD progression, ESRD, and CV events in diabetics with CKD. Use if eGFR ≥20.
- **GLP-1 Receptor Agonists (semaglutide, liraglutide, dulaglutide):** Reduce albuminuria and slow CKD progression. Cardiovascular benefits in diabetes.
- **Finerenone (non-steroidal MRA):** Add if albuminuria persists on maximal ACEi/ARB. Reduces CKD progression and CV events (FIDELIO-DKD trial).
- **Glycemic control:** Target A1c ~7% for most patients. Avoid hypoglycemia in advanced CKD (adjust insulin/sulfonylureas).
- **BP goal:** <130/80 mmHg.

PEARL: Modern "quadruple therapy" for DKD: **ACEi/ARB** + **SGLT2i** + **GLP-1 RA** ± **finerenone**.

Acute Kidney Injury (AKI)

Pre-renal (Hypoperfusion)
Clues: **BUN:Cr > 20:1**, FeNa < 1%
Causes: Dehydration, shock, CHF, cirrhosis.
Next: Fluid resuscitation (if appropriate).

Intrinsic (ATN)
UA: **"Muddy brown" casts**
Clues: FeNa > 2%
Causes: Ischemia, contrast, aminoglycosides.

Intrinsic (AIN)
Clues: Rash, fever, eosinophilia.
UA: **WBC casts**, eosinophils.
Causes: NSAIDs, PPIs, antibiotics.

Post-renal (Obstruction)
Imaging: Hydronephrosis on US.
Clues: Anuria or fluctuating output.
Causes: BPH, stones, tumors.